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An optimal federated learning-based intrusion detection for IoT environment

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Federated Learning (FL) allows the learning models in distributed systems to be trained by sharing the network data and model parameters. The attack patterns of attackers are frequently upgraded as well as the technology improves. Machine learning-based intrusion detection is familiar for cybersecurity in IoT networks. However, these traditional procedures mainly focus on training the machine learning model through specific data and parameters. This might reduce the detection performance of IDS as the system doesn't have insightful knowledge about the new attack patterns. Analyzing and detecting intrusions by analyzing diverse attack patterns is complex for machine learning algorithms. To overcome this, a federated learning-based intrusion detection approach is presented in this research work that trains deep learning classifiers in IoT networks through federated learning and detects different attacks. The Chimp optimization algorithm is used in the proposed work to select optimal features. Experimentations using the benchmark MQTT dataset validate that the FL-based IDS proposed in this research provides a maximum detection accuracy of 95.59% in detecting intrusions over traditional machine learning algorithms.

Keywords Federated learning, Deep learning, Intrusion detection, Network data, Security, Internet of things (IoT)

Over a decade, machine learning algorithms have been considered as an effective mechanism to detect intrusions in a network. Numerous ML-based IDS are developed to detect intrusions in IoT networks and other wireless sensor networks. However, the major limitation of ML-based IDS is its large and diverse dataset requirement for training the network. Practically providing a large dataset for ML models to train the network is challenging, and this challenge has increased in recent times due to dynamic, distributed heterogeneous networks. Besides this, ML models face privacy and security issues due to their centralized characteristics. The chances of data leakage, data breach, and unauthorized access decrease the confidence in adopting ML in data-sensitive applications[1]. To overcome these challenges, Federated Learning (FL) has been presented recently to enable distributed learning features without compromising security and privacy factors. The inherent heterogeneity and vast volume of IoT-generated data exacerbate these limitations, making it increasingly difficult for centralized ML models to adapt to evolving attack patterns and maintain data privacy. These challenges underscore the need for decentralized, privacy-preserving, and scalable solutions. To address these issues, the proposed framework introduces a Federated Learning (FL)-based IDS that leverages the collaborative power of FL to train models across distributed devices without compromising sensitive data. By integrating a novel Chimp Optimization Algorithm for feature selection, the framework ensures reduced dimensionality of network data and improved classification accuracy. This dual combination of FL and optimized feature selection introduces a unique approach to enhancing both detection performance and data privacy in IoT environments. The novelty of the proposed framework lies in its ability to efficiently handle complex attack scenarios, including Distributed Denial-of-Service (DDoS) and brute force attacks, by combining decentralized learning, advanced optimization, and deep learning classifiers. These contributions position the proposed model as a significant advancement in intrusion detection for IoT networks.

Using FL, multiple clients can train a common model without disclosing private information[2]. The model can be trained by each client locally using their own data without sharing it with the central server. The client's model parameters are shared with central servers. The FL aggregator aggregates the client model updates and provides a global model to the client for further processing. The clients replace the current model with a global

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model to enhance the detection performance[3]. Due to this, the scalability and efficiency of IDS are improved in the client networks.

Recently various FL-based IDS have been developed by the research community, and most of the models perform better than traditional machine learning-based IDS[4]. Traditional Intrusion Detection Systems (IDS) face significant challenges in IoT environments due to the decentralized nature of these networks and the volume of data generated by interconnected devices. Centralized machine learning models are commonly used to detect cyberattacks but it requires large datasets and involve privacy risks as sensitive information must be transmitted to a central server for analysis. Deep learning models are widely preferred over ML models in various domains[5,6]. Similarly, some of the deep learning models perform better in detecting different types of intrusions[7]. However, to provide a decentralized approach to detecting intrusions and experiencing different types of attack patterns, the FL-based IDS are popular among network environments. The features from the network data are selected directly by the deep learning models, which might increase the complexity of the classification process. To address these challenges, this research proposes a Federated Learning-based Intrusion Detection System (FL-IDS) that utilizes the features of Federated Learning (FL) and the Chimp Optimization Algorithm. FL allows multiple IoT devices to collaboratively train a model without sharing raw data which enhances the privacy. The Chimp Optimization Algorithm is applied to perform feature selection. The essential features from the network traffic data are selected by the optimization algorithm thus, it will reduce the complexity and increase the detection accuracy of the classifier. Moreover, due to the updated global parameters, all the clients will experience this improved detection accuracy. The objective of proposed technique is to handle the complex cyberattacks. Since IoT networks are frequently targeted by multiple attacks such as distributed denial-of-service (DDoS), brute force, and data poisoning attacks. These attacks introduce vulnerabilities in distributed systems and traditional IDS approaches struggle to handle this evolving threat patterns. By distributing the learning process across devices and selecting the most critical features the proposed optimized FL-IDS for IoT network provides enhanced detection accuracy and ensures privacy. The research work made the following contributions.

- A decentralized Federated Learning-based Intrusion Detection System (FL-IDS) is presented for IoT networks. The proposed method improves data privacy and security by allowing local devices to train models on their own data and share only the model parameters with a central server for aggregation which secures the sensitive information.
- Proposed a novel optimal feature selection model using Chimp Optimization in the IDS domain to ensure efficient feature selection. This reduces the dimensionality of network traffic data and enhances classification accuracy.
- The remote IoT instances are provided with convolutional neural network-based IDS with different parameter settings. Further, through FL-aggregation, the parameters are aggregated to create a global model and shared with clients to attain improved detection accuracy in all instances.
- Detailed experimental evaluation of the proposed model is presented a using a benchmark MQTT dataset, which has different IoT device traffic data. The Proposed model performance is evaluated and compared with conventional ML models like decision tree, random forest, neural network, gradient boost and naïve bayes algorithms. With a detection accuracy of 95.59% the proposed model performs better than traditional machine learning-based IDS approaches. This highlights the proposed model ability in effectively detect multiple types of intrusions in IoT environments, including advanced attacks like DDoS and brute force attacks.

The discussions in the following sections are arranged as: Sect. 2 provides a detailed literature review of recent research works on IDS. Section 3 presents the proposed FL-based IDS. Experimentations using benchmark dataset and the results are presented in Sect. 4. The conclusion is presented in Sect. 5.

Related works

Recent research works on Intrusion detection in IoT environments are presented in this section. Selecting suitable ML model for malicious traffic classification is an important process. The review work presented in[8] analyzes various ML model performances which are suitable for smart city traffic classification. selecting suitable ML to handle the cyber-attacks in smart city is addressed in[9]. The presented model selects the optimal features from network data and then utilizes ML algorithms to classify the attacks. Machine learning-based IDS faces difficulty in detecting intrusions due to the enormous amount of data and distributed network-connected devices. By addressing this, an FL-based IDS was proposed in[10], which shares local parameters with a central server and aggregates to generate a global model. Based on the global model parameters, the clients are modified and trained with local data to detect intrusions in the IoT network. Experimentation using the Edge IIoT dataset attained an accuracy of 92.49%, which is slightly higher than ML-based approaches.

The FL-based IDS presented in[11] includes a discriminator network for attack detection in IoT edge devices. Using deep learning model, feature extraction is performed and through discriminator network the features are classified. Reduced incorrect prediction and improved security are the features of the presented FL-IDS model. The FL based IDS presented for Industrial IoT environment addresses the heterogeneity issues in the FL aggregation[12]. The detect attacks in the client instances an unsupervised device clustering procedure was presented in this research which detects the presence of various attackers in IIoT network. For DDoS attack detection, a federated integrated network IDS was presented in[13] which analyzes the anomalous behavior through HIDS and detect attacks in the network. Additionally using federated learning, the local system parameters are updated to detect the anomalous traffic. Using this federated learning procedure, the detection accuracy was improved and privacy issue in decentralized infrastructure is reduced.

The IDS presented in[14] utilizes FL to share the weights of local models and update with global model parameters in IoT environment. The presented Federated averaging procedure utilizes different learning algorithms in client

locations and shares the parameters to central server. Based on the global model, the local model parameters are updated to detect intrusions. Experimentations utilizes ToN-IoT dataset validates the better detection accuracy of presented model over traditional approaches. The IDS presented in[15] includes federated learning and active learning techniques to detect network flow-based intrusions in an IoT environment. The federated learning model used to train the models in federation and the semi-supervised active learning model is used for global model adaption. Experimentations using benchmark dataset validates that due to global model parameter update procedure, the detection performance is increased by 7% compared to traditional procedure.

The FL based IDS presented in[16] combines anomaly-based IDS with FL to ensure the privacy of IIoT networks. The presented model adopts instance-based transfer learning procedure at the local systems and includes a rank aggregation with weighted voting approach for detecting intrusions with improved accuracy over Ada boost and random forest algorithms. The IDS presented in[17] aimed to improve the detection accuracy and network efficiency by detecting different attacks through federated learning procedure. The presented anomaly IDS computes an intrusion score for the normal traffic using different variants of autoencoder models. Based on the score, the intrusions are detected with better accuracy. Experimentations using conventional IDS2017 and 2018, TFC2016 datasets validates the proposed model performance through various metrics. A simple federated sampling and light weight IDS presented in[18] samples the network traffic using k-means and detects intrusions in semi supervised learning procedure. The data privacy is ensured in the presented model through local clustering so that summary statistics are shared to central server instead of privacy information.

To ensure the robustness and balance the performance in detecting intrusions in IoT networks an FL-based NIDS was presented in[19] which utilizes federated artificial intelligence techniques to provide horizontal scalability and data confidentiality. In the presented approach the models are initially trained with gaussian noise values and then black box attacks are performed. Finally, the attacks are detected learning models with better accuracy. Experimentation presents that the proposed IDS has better detection performances even for different sizes of data samples. The FL based IDS presented for agricultural IoT networks prevents attacks by training the deep learning algorithms through the global parameters[20,21]. Recurrent neural network, deep neural network, and convolutional neural network are employed in the remote locations and their parameters are aggregated and updated with global parameters. Experimentations using benchmark datasets presented the better performance of FL model. However, the classifier computation complexity is comparatively higher in the presented model.

The FL-based IDS presented in[22] utilizes gated recurrent units in client locations and shares the parameters to FL-aggregation module to generate a global model. The learned weights are updated with the client IDS and detects intrusions with better accuracy over traditional ML based approaches. Similar GRU with FL based IDS presented in[23,24] which updates the learning models parameters based on global model parameter provided by the central server. The presented model experimentation shows that the detection performance is improved by 8% over traditional approach and reduces the communication cost in IoT environment compared to other FL algorithms. The anomaly and malicious traffic detection model presented in[25] for an IoT network utilizes machine learning algorithm with a novel feature selection model. The presented feature selection model utilizes wrapper techniques to filter the features and selects the optimal features using ML algorithm area under the curve metric. Finally using Shannon entropy based softest the detection model validates the selected features in the malicious traffic identification process. similar model presented in[26] utilizes wrapper technique for feature filtering and utilizes different ML classifiers to detect different intrusions and malicious events in an IoT traffic. The IDS presented in[27] combines FL with LSTM to detect the network intrusions in IoT environment. The presented approach deploys LSTM global model to all users and train the model. After training the parameters are updated with central server and based on the aggregated parameters a global model is generated and updated with clients. Due to this, the detection accuracy of client IDS is improved better than traditional detection procedures. IoT anomalies have also been extensively research in the literature[28,29].

From the above study, it can be summarized that the role of learning algorithms specifically deep learning algorithms is inevitable in IDS designs. However, by combining FL with deep learning the performance can be improved further and security issues can be reduced. In few research works for feature extraction deep learning models are utilized. Including optimization model for better feature processing will enhance the intrusion detection performance. Thus, in this research work an optimal FL-based IDS was presented to for intrusion detection in an IoT environment.

Proposed work

The proposed optimal FL based IDS for IoT Environment aimed to detect different intrusions in IoT environment. The complete overview of the proposed FL-based ID model is presented in Fig. 1. The high-level architecture includes multiple IoT instances distributed at different sites. All the IoT instances are connected through gateways, and the network data will be shared through the gateway. An intrusion detector will be placed in the security gateway for monitoring the IoT instance traffic data. This makes the architecture more independent and intelligent so that the system does not need to rely on traditional server-device mechanisms and procedures for decision-making. Also, this intelligent detection mechanism reduces the detection time by analyzing the shared network data. Federated setup is introduced in the gateway arrangement. For every stipulated time, the learning models used in the IoT instances will be trained on local network data. Then, the model parameters are aggregated to create a global model so that more sophisticated detection performances can be obtained for all the models. The main feature of FL in the research work is to share the learning models with server and aggregating the learning models will provide a global intelligence model which can able to detect the attacks with higher accuracy than traditional modes. The global optimized model used in the gateways increases the knowledge of gateways about empowers IoT instances to detect intrusions better manner.

The proposed architecture for the Federated Learning-based Intrusion Detection System (FL-IDS), as shown in Fig. 1, is designed to effectively address key challenges in IoT environments, such as scalability, data privacy,

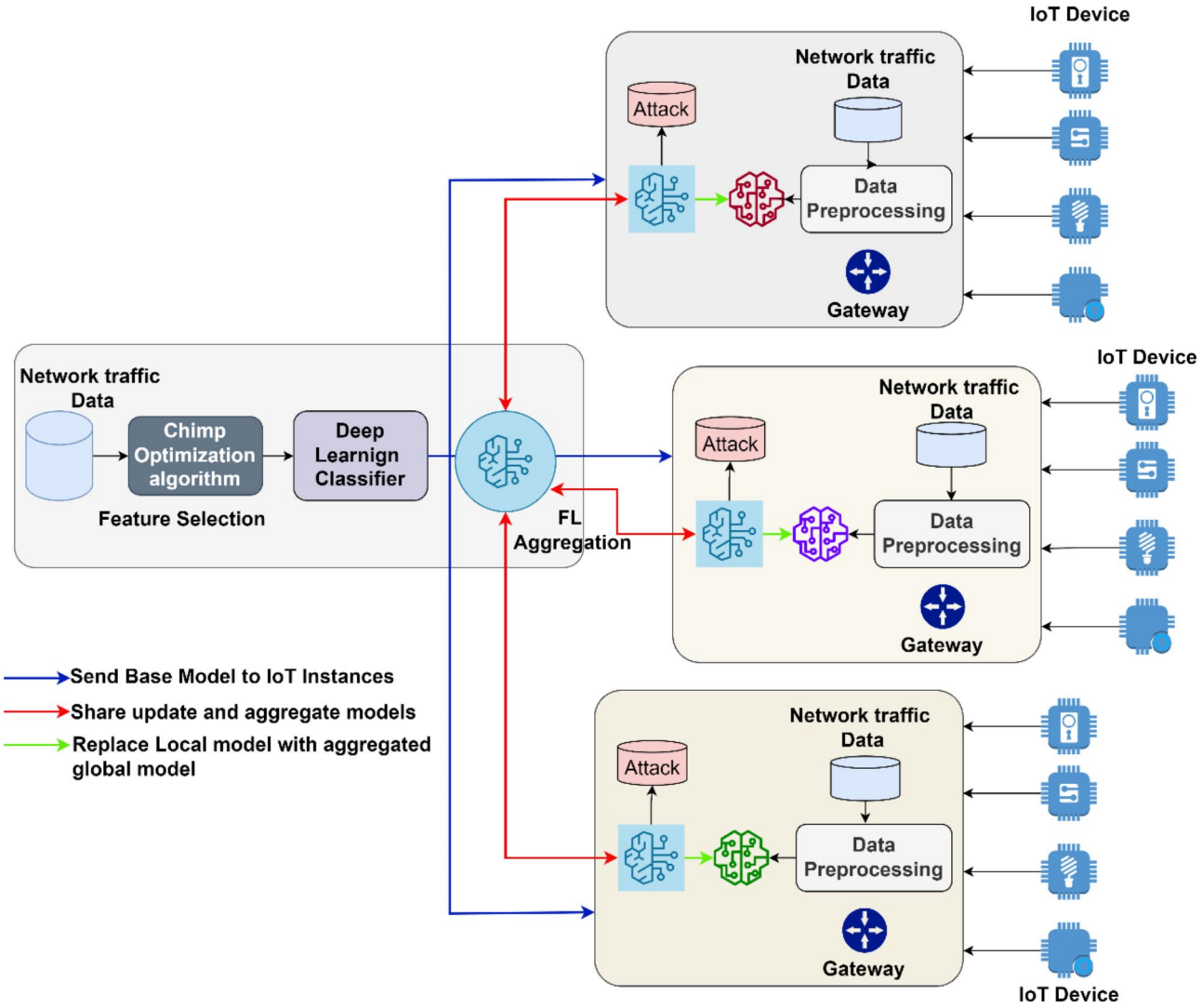


Fig. 1. Proposed Optimal FL-based IDS architecture.

and computational efficiency. At its core, the architecture is built around distributed IoT instances, which act as data generators and execute local intrusion detection using trained Convolutional Neural Network (CNN) models. These instances enable decentralized processing, ensuring data privacy by keeping raw data within local devices and significantly reducing the risk of data breaches. Security gateways serve as intermediaries between the IoT instances and the central server. These gateways play a crucial role in managing the data flow, hosting the feature selection module, and executing localized intrusion detection tasks. By processing data at the gateway level, the architecture reduces the reliance on centralized systems, lowering latency and enhancing fault tolerance. The feature selection module, powered by the Chimp Optimization Algorithm, optimizes the dataset by identifying and selecting the most relevant features from network traffic data. This step is essential for reducing the dimensionality of the data, improving classification accuracy, and minimizing the computational load, especially in resource-constrained IoT devices.

Each IoT instance is equipped with a local CNN-based learning model that processes the optimized feature set for training. These local models are periodically updated and shared with a Federated Aggregation Server. The server aggregates the parameters from all local models to create a global model, which is redistributed back to the IoT instances. This global model ensures consistency across the system and enhances the detection of evolving attack patterns, such as Distributed Denial-of-Service (DDoS) and brute force attacks. The integration of Federated Learning with optimization ensures privacy-preserving data processing while maintaining high detection accuracy. This innovative combination makes the architecture scalable, efficient, and adaptable to heterogeneous IoT environments, distinguishing it from traditional IDS solutions.

Preprocessing the network data is the first step of the proposed ID model. The source and destination connection has TCP packet sequences. Each packet will have different features. Network logs information will be present in each packet and the bidirectional features are used to detect the intrusions. Next to preprocessing, feature selection is performed where Chimp Optimization Algorithm is used in the proposed model. Due to the feature benefits of fast convergence and doesn't require large computational resources to find optimal solution,

chimp optimization will suit better than other optimization algorithms in the feature selection process of ID model. The thinking ability, hunting behavior, and sexual motivation of chimps are formulated in the optimization algorithm. The ability to find optimal solution in complex environment is better in chimp optimization thus it is used in the proposed model for feature selection.

The four stages of chimp optimization are driving, attacking, chasing, and blocking. The optimization model was initialized randomly with chimps. The prey breakout path was predicted by the attacker and it forces the prey to move close to chaser. The prey was followed by driver without performing any attack and barrier chimp generates barriers in the attack process. The process of chasing and driving is mathematically formulated as

$$d = |c.x_{prey}(r) - m.x_{chimp}| \tag{1}$$

$$x_{chimp}(r+1) = x_{prey}(r) - a.d \tag{2}$$

$$a = 2fr_1 - f \tag{3}$$

$$c = 2fr_2 \tag{4}$$

where the prey location vector is indicated using x_{prey} and the chimp location vector is indicated using x_{chimp}. The parameters a, c, and m are the coefficient vectors and the parameter r_1 and r_2 are random vectors and its range are given as [0,1] and [0,2.5] respectively.

In the hunting process, chimps search for prey locations and then evaluate the prey position based on their hunting strategy. The chimps in the group update the position of prey, and this hunting process is mathematically given as

$$d_{attacker} = |c_1.x_{attacker} - m_1.x_1| \tag{5}$$

$$d_{barrier} = |c_2.x_{barrier} - m_2.x_2| \tag{6}$$

$$d_{chaser} = |c_3.x_{chaser} - m_3.x_3| \tag{7}$$

$$d_{driver} = |c_4.x_{driver} - m_4.x_4| \tag{8}$$

where $x_1 = x_{attacker} - a_1(d_{attacker})$ (9)

$$x_2 = x_{barrier} - a_2(d_{barrier}) \tag{10}$$

$$x_3 = x_{chaser} - a_3(d_{chaser}) \tag{11}$$

$$x_4 = x_{driver} - a_4(d_{driver}) \tag{12}$$

$$x(t+1) = \frac{x_1 + x_2 + x_3 + x_4}{4} \tag{13}$$

where $x(t+1)$ indicates the chimp updated location, $x_{attacker}$ indicates the first better search agent, $x_{barrier}$ and x_{chaser} are the second and third better search agent. The fourth better search agent is x_{driver}. After hunting, the sexual motivation of chimps is mathematically formulated as

$$x_{chimp}(t+1) = \begin{cases} x_{prey}(l) - a.d & if\ \delta < 0.5 \\ choatic_value & if\ \delta > 0.5 \end{cases} \tag{14}$$

where δ is a random number with in the range [0,1]. In real, chimps will continuously change their location, thus in the proposed model feature selection process, the fitness of chimps are formulated as a function to select the essential features and to improve the detection accuracy. The fitness function is mathematically formulated as

$$Fit_{chimp}(i) = \alpha . A(i) + (1-\alpha) . log_{10}(N/n(i)) \tag{15}$$

where $Fit_{chimp}(i)$ indicates the ith chimp fitness function and the total accuracy of ith chimp is indicated using $A(i)$. N defines the total number of features and $n(i)$ is the features selected by i^{th} chimp. The weighted parameter α is assumed to be less than one, and its optimal value has been obtained by train and error procedure.

Next to optimal feature selection, the architecture has learning models. In the proposed work convolutional neural network is used for training procedures. The CNN model has multiple convolution and pooling layers and provides classification outputs which will be used to analyze the intrusions. The CNN model will be trained with different patterns of attacks and the final SoftMax function classify the network traffic into different attack categories. The optimal features selected by the optimization models are given as input to the deep learning classifier. The temporal information in the feature sets is learned by the CNN and get trained to detect the attacks. Once the model is trained, the trained CNN model will be shared among other IoT instances for detecting attacks in the real time network data of IoT devices.

The federated learning procedure is associated will all the IoT instances. Each instance connected in the architecture will have a training data and using the shared view of learning model the classifier in the instance gets trained. Once it gets trained, the parameters are shared with the central server for aggregation. The process of federated learning is simplified in the following steps.

Step 1: Share the initial model with other IoT instances.

Step 2: Based on the base model from server, the client learning model will get retrained using the local network data.

Step 3: Feature information in the network data is provided as input to the learning model for attack detection. Using the training parameters, *learning models detects the attack classes.*

Step 4: After training the learning model for several epochs, *the classification model will be sent to server for FL aggregation. Based on all the classification model parameters, a global model will be created.*

Step 5: The created global model is shared with other IoT environment. In client side, *the current model is replaced with global shared model and trained with local data to detect attacks.*

Due to FL aggregation, the communication overhead is reduced as the sharing procedure utilizes only the parameter details, not any other sensitive information from local data. Mathematically, the FL aggregation process is formulated as

$$F_k(w) = \frac{1}{n_k} \sum_{i \in p_k} f_i(w) \tag{16}$$

$$f_i(w) = \sum_{k=1}^{K} \frac{n_k}{n} F_k(w) \tag{17}$$

where the number of clients or IoT instances are indicated using K and its index is indicated using k. The total number of data samples used to train the instance k is given as n_k. The proposed model attack detection procedure is summarized in the pseudocode given below.

Input: Train and test the Network data
Output: Intrusion type
Initialize the IoT instances $d = d_1, d_2, .., d_n$
Initialize chimp optimization
while max_iterations
    Hunt the prey based on hunting strategies of $d_{attacker}$, $d_{barrier}$, d_{chaser} , d_{driver}
        Update position of prey and chimp
            Calculate the fitness using Fit_{chimp} (i)
    End function
While d_i in d do
    Train (data)
        Return
    End
FL average
Update parameters to other IoT instances
Train using global parameters
Update attack detection patterns
End
End

Algorithm. Optimal FL based IDS for IoT environment.

Results and discussion

For the proposed model evaluation, the experimentation utilizes MQTT protocol dataset which is available in Kaggle repository[30]. This dataset simulates real-world IoT device communications and contains both normal and attack traffic. The attack types include.

- Brute Force Attack - A type of cyberattack where multiple login attempts are made to gain unauthorized access.
- Flooding DoS (Denial of Service) - An attack that overwhelms the network with excessive data, leading to disruption.
- MQTT Publish Flood - A targeted attack on the MQTT protocol, causing a flood of publish requests to overwhelm the system.
- Malformed Data Attack - An attack that involves sending corrupted or unexpected data to exploit system vulnerabilities.
- SlowITe Attack - An advanced attack that uses slow HTTP connections to exhaust server resources.

Attack	Number of Packets	Packet size (Mb)
Flooding DoS	130,223	49.87
MQTT Publish flood	613	8.212
SlowITe	9202	9.722
Malformed data	10,924	1.038
Brute Force	14,501	1.397

Table 1. MQTT dataset description.

Models	Parameter	Range/Type
Neural network	Number of layers	3
	Number of Neurons per layer	128,64, 32
	Learning Rate	0.01
	Activation function	ReLU
Random Forest	Number of estimators	100
	Max depth	20
	Min sample split, leaf	2,1
	Criterion	Gini
Naïve bayes	Alpha	1.0
Decision tree	Max depth	10
	Min sample split, leaf	2,1
	Criterion	Gini
Gradient boost	Number of estimators	20
	Learning Rate	0.1
	Max depth	3
MLP	Batch size	1000
	Activation	ReLU
	Optimizer	Adam
	Learning Rate	0.001
	Hidden layers	100,50
	Regularization term (L_2)	0.001
Proposed FL-CNN	Batch size	1000
	Number of Filters in conv layer	32,64,128
	Convolution Kernel size	3 × 3
	Drop out	0.5
	Activation	ReLU
	Optimizer	Adam

Table 2. Simulation hyperparameters.

The experimentation was performed using python tool with essential libraries and tools. The PyTorch library was used for implementing the deep learning model which serves as the classifier in the proposed model. SciPy and NumPy libraries are used for mathematical and computational operations in optimization model. The experimental setup of proposed model has an Intel i5 processor with 32GB memory and the simulation tool used for evaluation is Python. Table 1 presents the details about the dataset. Before training and testing the dataset is balanced and in the balanced data 70% of the data are used for training and 30% is used for testing.

In the experimentation process, the data in the MQTT dataset is processed initially by the Chimp Optimization Algorithm to reduce the dimensionality of the dataset by selecting the most relevant features from network traffic data. This step ensures that the classifier receives only the most critical information and reduces the computational complexity and increases the accuracy. After optimal feature selection, in the local model training, each IoT instance in the federated learning setup trained its own CNN model using local data. Further in the Federated Aggregation, the trained models from each instance were sent to the central server and the parameters are aggregated to form a global model. Finally, to test and evaluate the model, the global model is then tested and performance metrics such as accuracy, precision, recall, and F1-score are computed. The proposed model performance is comparatively evaluated with existing machine learning models like Multilayer perceptron, Decision tree, Random Forest, Neural network, Gradient boost, Naïve bayes algorithms[31]. The simulation parameters of the existing machine learning algorithms and the proposed model is presented in Table 2.

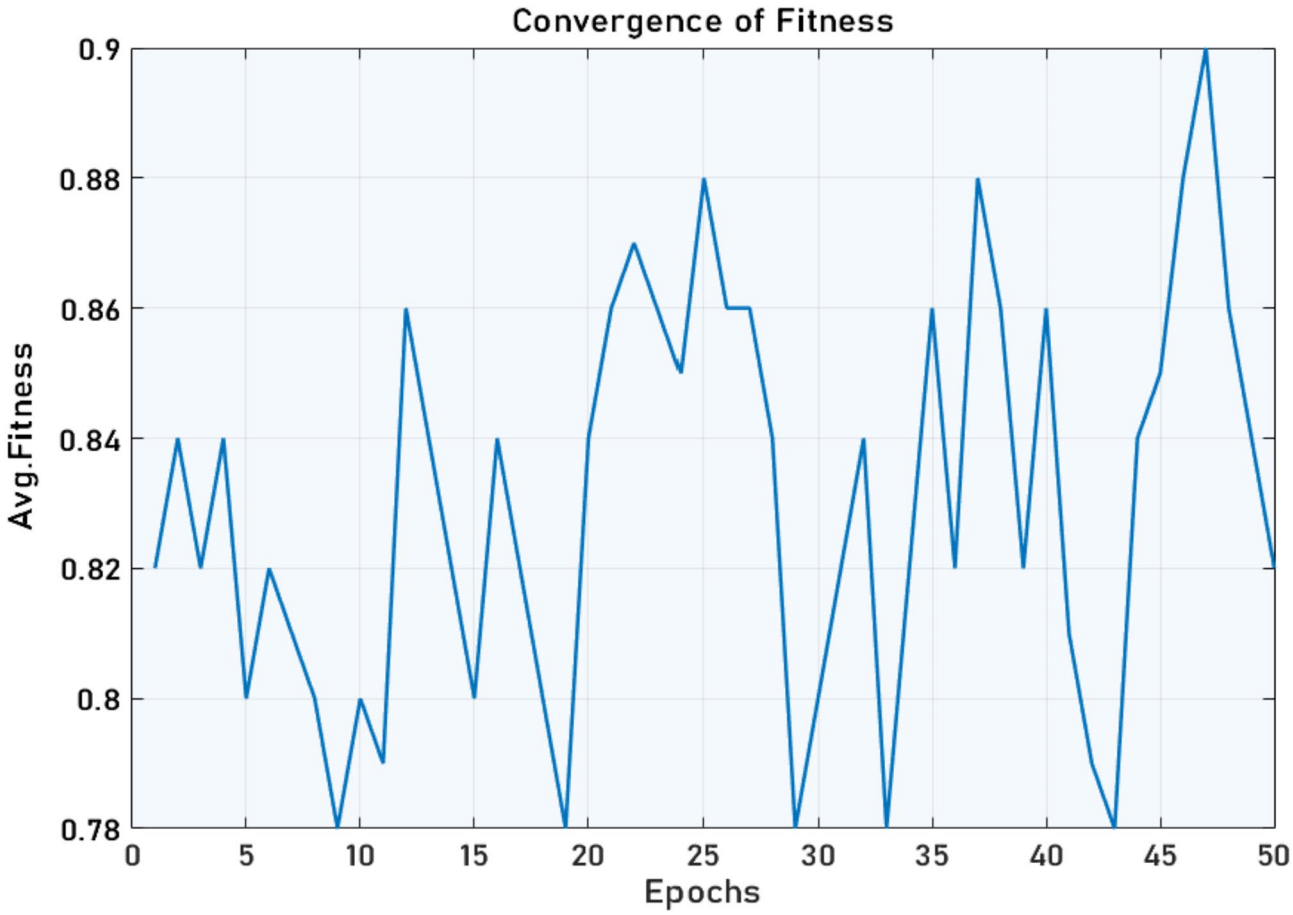


Fig. 2. Optimization convergence curve.

tcp.flags, tcp.time_delta, mqtt.conack.flags, mqtt.conack.flags.sp, mqtt.conflag.qos, mqtt.conflags, mqtt.dupflag, mqtt.hdrflags, mqtt.len, mqtt.msgtype, mqtt.protoname, mqtt.sub.qos, mqtt.suback.qos, mqtt.ver, mqtt.willmsg, mqtt.willmsg_len, mqtt.willtopic_len

Fig. 3. List of Features selected by the optimization model.

The metrics used for proposed model performances evaluation are recall, precision, f1-score and accuracy. The optimization model performance is evaluated, and the list of optimal features selected by the optimization model is presented in Fig. 2. The convergence curve indicates the proposed optimized feature selection process, selects the optimal features with better average fitness function. Out of 34 generalized features, the proposed optimized feature selection model, selects 17 essential features which are used to detect the intrusions in the IoT network. The list of selected features is presented in Fig. 3.

The existing and proposed model performances is evaluated through the confusion matrix obtained during the test process. For each model, the confusion matrix is obtained and the metrics are calculated so that it can be used to compare with the proposed model.

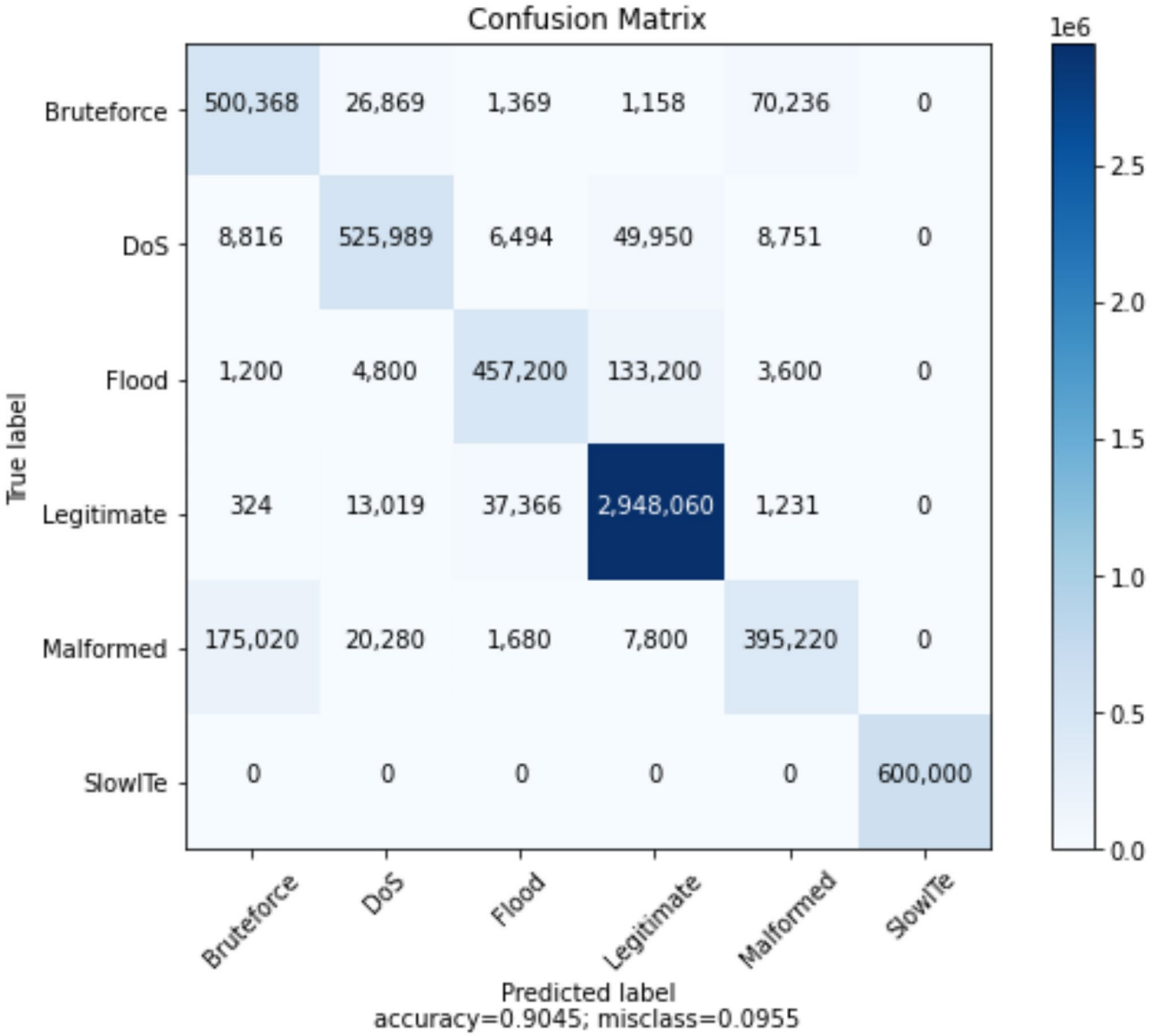


Fig. 4. Neural Network model confusion matrix.

Class	Recall	Precision	F1-Score	Accuracy
Brute Force	0.8339	0.7297	0.7783	0.9045
Dos	0.8766	0.8901	0.8833	
Flood	0.762	0.9069	0.8282	
Legitimate	0.9827	0.9388	0.9603	
Malformed	0.6587	0.825	0.7325	
SlowITe	1	1	1	

Table 3. Neural network model performance metrics for each class.

Figure 4 depicts the confusion matrix for Neural network model which shows that the model has attained 0.9045 as classification accuracy with misclassification rate of 0.0955. The class recall, precision, and f1-score attained by the neural network model is presented in Table 3 and from that, the average precision value is obtained as 0.8818, recall as 0.8523 and f1-score as 0.8668.

Figure 5 depicts the confusion matrix for random forest model and it shows that the model has attained 0.9160 as classification accuracy with misclassification rate of 0.0840. The class precision, recall and f1-score

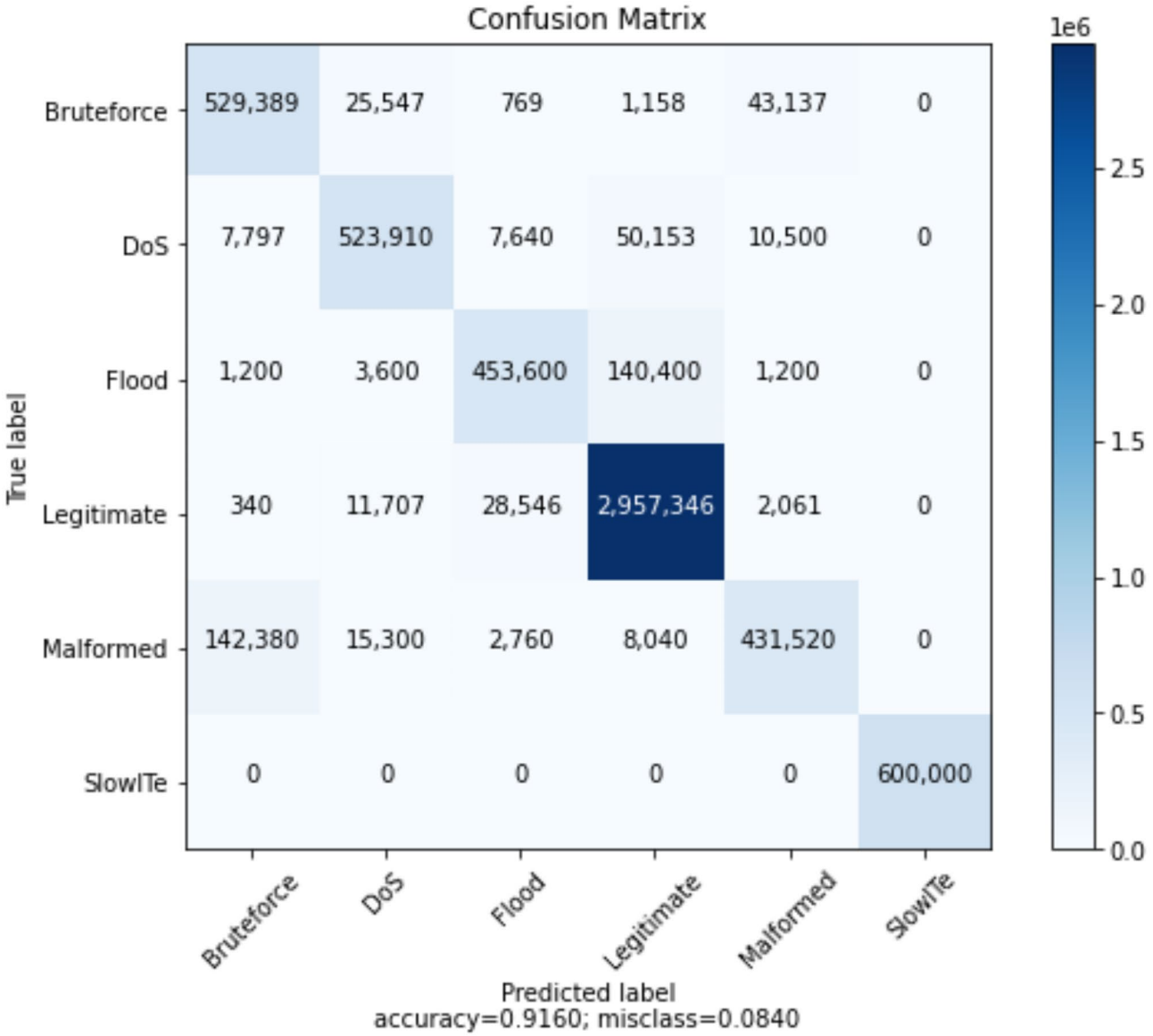


Fig. 5. Random Forest model confusion matrix.

Class	Recall	Precision	F1-Score	Accuracy
Brute Force	0.8823	0.7772	0.8265	0.916
Dos	0.8732	0.9032	0.8879	
Flood	0.756	0.9195	0.8298	
Legitimate	0.9858	0.9367	0.9606	
Malformed	0.7192	0.8835	0.7929	
SlowITe	1	1	1	

Table 4. Random forest model performance metrics for each class.

attained by the neural network model is presented in Table 4 and from that, the average precision value is obtained as 0.9034, recall as 0.8694 and f1-score as 0.8861.

Figure 6 depicts the confusion matrix for Naïve Bayes model and it shows that the model has attained 0.6439 as classification accuracy with misclassification rate of 0.3561. The class precision, recall and f1-score attained by the neural network model is presented in Table 5 and from that, the average precision value is obtained as 0.8427, recall as 0.6588 and f1-score as 0.7395.

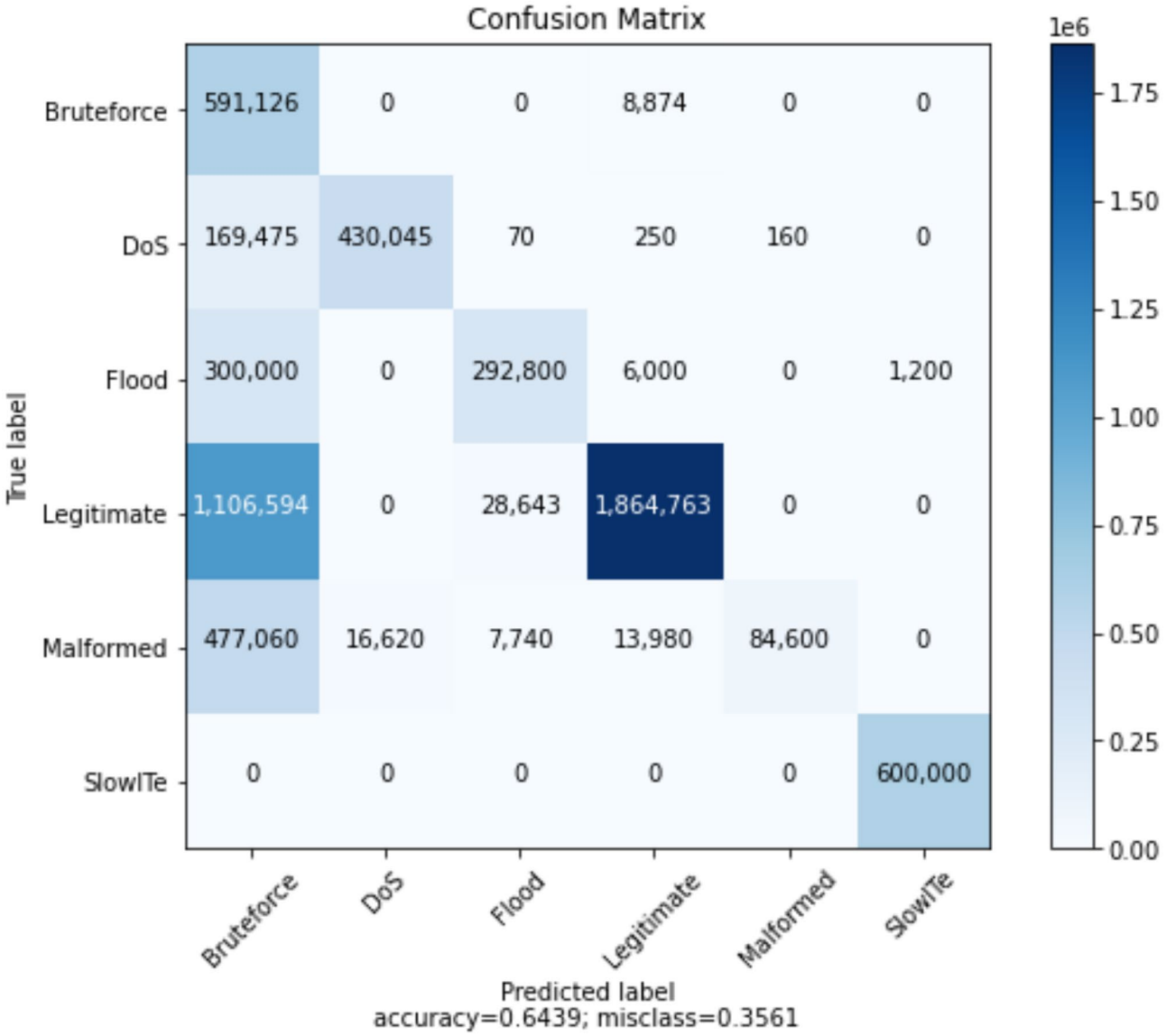


Fig. 6. Naïve Bayes model confusion matrix.

Class	Recall	Precision	F1-Score	Accuracy
Brute Force	0.9852	0.2236	0.3644	0.6439
Dos	0.7167	0.9628	0.8217	
Flood	0.488	0.8893	0.6302	
Legitimate	0.6216	0.9846	0.7621	
Malformed	0.141	0.9981	0.2471	
SlowITe	1	0.998	0.999	

Table 5. Naïve Bayes model performance metrics for each class.

Figure 7 depicts the confusion matrix for MLP model and it shows that the model has attained 0.9038 as classification accuracy with misclassification rate of 0.0962. The class precision, recall and f1-score attained by the neural network model is presented in Table 4 and from that, the average precision value is obtained as 0.8803, recall as 0.8519 and f1-score as 0.8629. Table 6 shows the MLP model performance metrics for each class.

Figure 8 depicts the confusion matrix for gradient boost model and it shows that the model has attained 0.8842 as classification accuracy with misclassification rate of 0.1158. The class precision, recall and f1-score attained by the neural network model is presented in Table 7 and from that, the average precision value is obtained as 0.8575, recall as 0.8264 and f1-score as 0.8416.

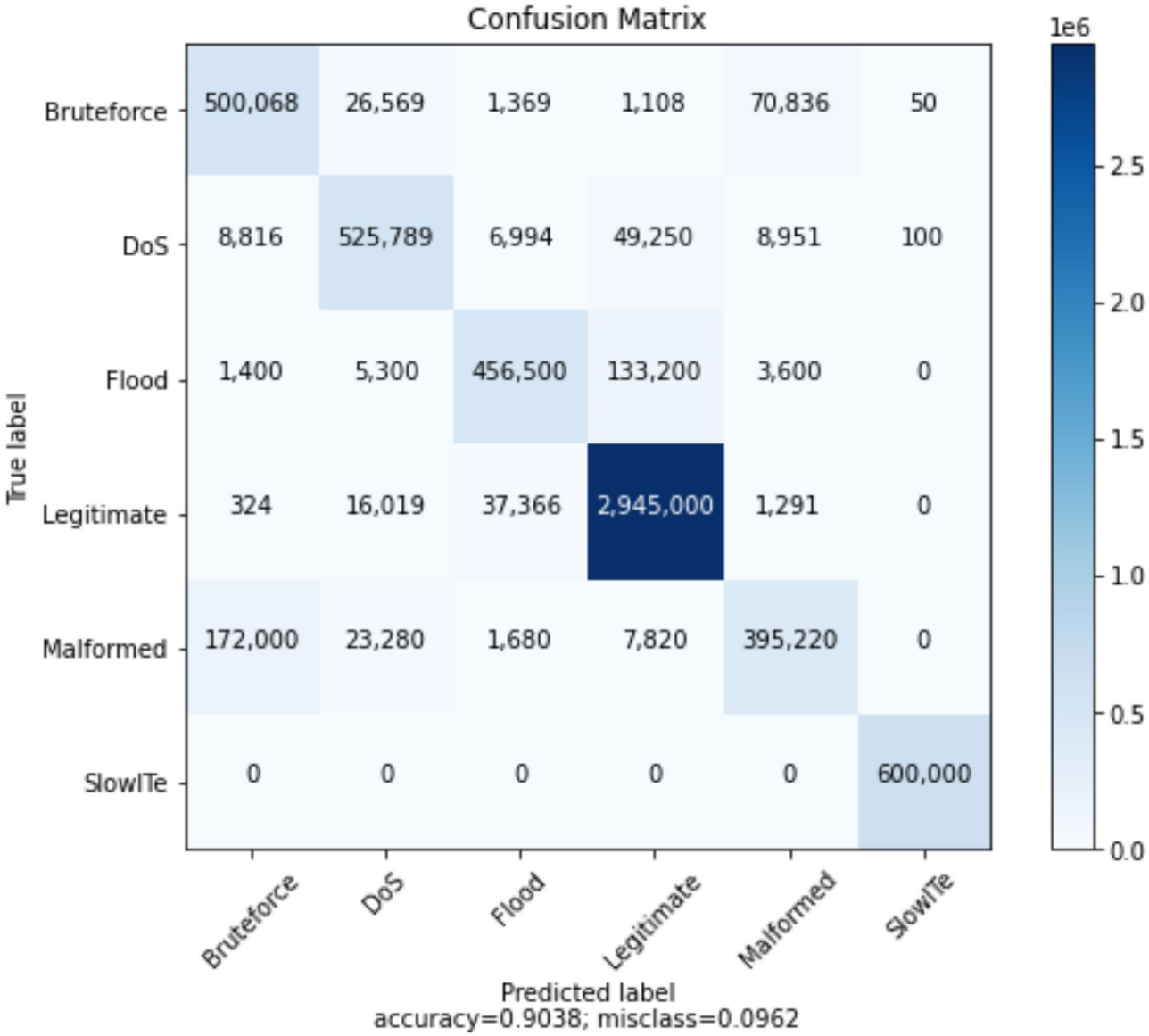


Fig. 7. MLP model confusion matrix.

Class	Recall	Precision	F1-Score	Accuracy
Brute Force	0.8334	0.7326	0.7798	0.9038
Dos	0.8765	0.8808	0.8786	
Flood	0.7608	0.9059	0.8271	
Legitimate	0.9817	0.939	0.9598	
Malformed	0.6587	0.8236	0.732	
SlowITe	1	0.9998	0.9999	

Table 6. MLP model performance metrics for each class.

Figure 9 depicts the confusion matrix for decision tree model and it shows that the model has attained 0.9159 as classification accuracy with misclassification rate of 0.0841. The class precision, recall and f1-score attained by the neural network model is presented in Table 4 and from that, the average precision value is obtained as 0.9033, recall as 0.8694 and f1-score as 0.8860. Table 8 represents the decision tree model performance metrics for each class.

Figure 10 depicts the confusion matrix for proposed FL-CNN model and it shows that the model has attained maximum accuracy of 0.9559 with misclassification rate of 0.0441. The class precision, recall and f1-

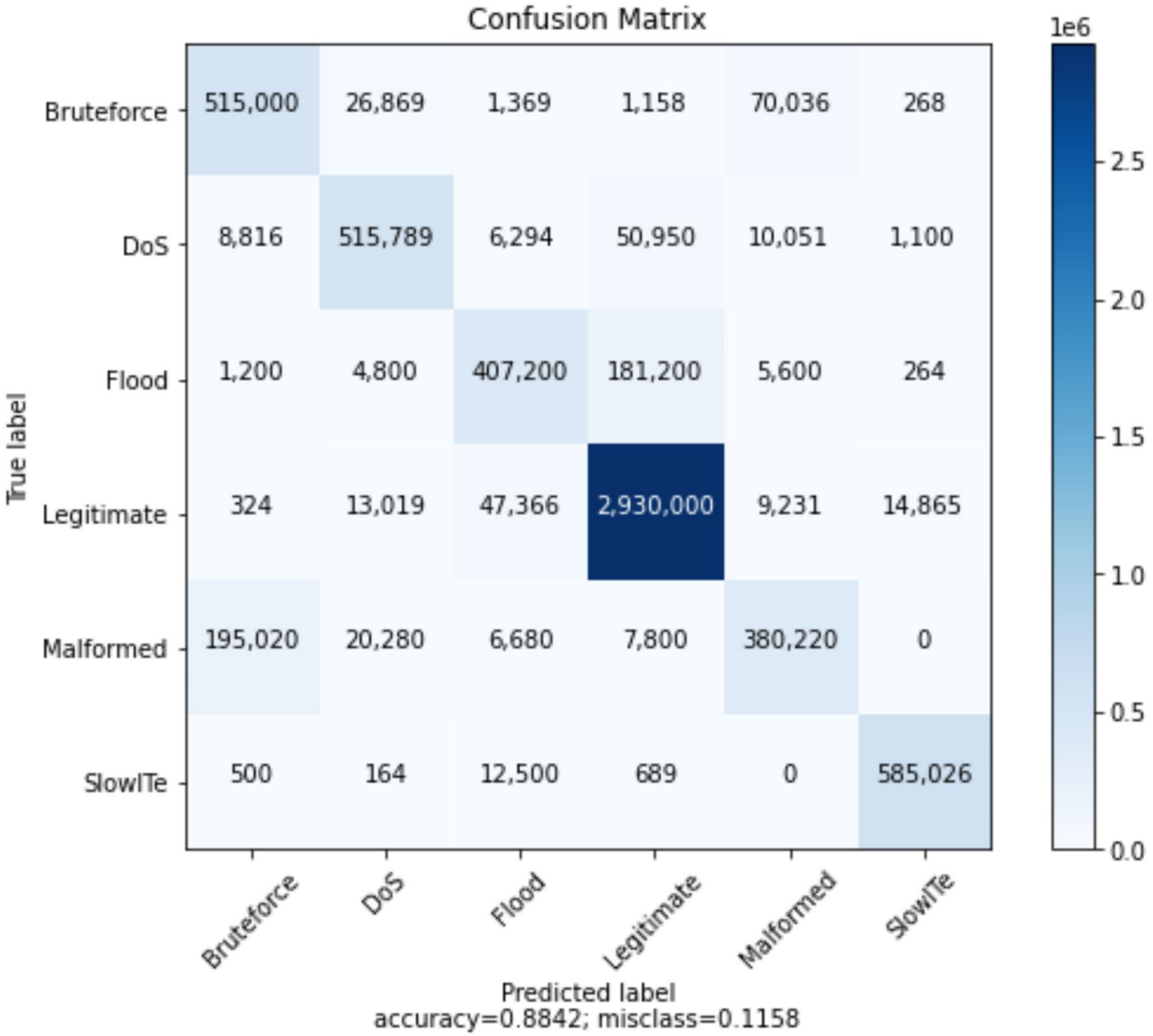


Fig. 8. Gradient Boost model confusion matrix.

Class	Recall	Precision	F1-Score	Accuracy
Brute Force	0.8378	0.7144	0.7712	0.8842
Dos	0.8698	0.8879	0.8787	
Flood	0.6784	0.8459	0.7529	
Legitimate	0.9719	0.9238	0.9472	
Malformed	0.6233	0.8002	0.7008	
SlowITe	0.9769	0.9726	0.9747	

Table 7. Gradient boost model performance metrics for each class.

score attained by the neural network model is presented in Table 9 and from that, the average precision value is obtained as 0.9479, recall as 0.9330 and f1-score as 0.9404.

The average values obtained by each algorithm are comparatively analyzed and the precision-recall metric comparative analysis is shown in Fig. 11. From the analysis, it can be observed that the performance of proposed FL-CNN is better than the existing algorithms. The proposed model average precision is 0.9479 which is 4.4% better than random forest and decision tree models, approximately 6.6% better than Neural network and MLP models. The proposed model attained precision is approximately 9% better than gradient boost and 10% better

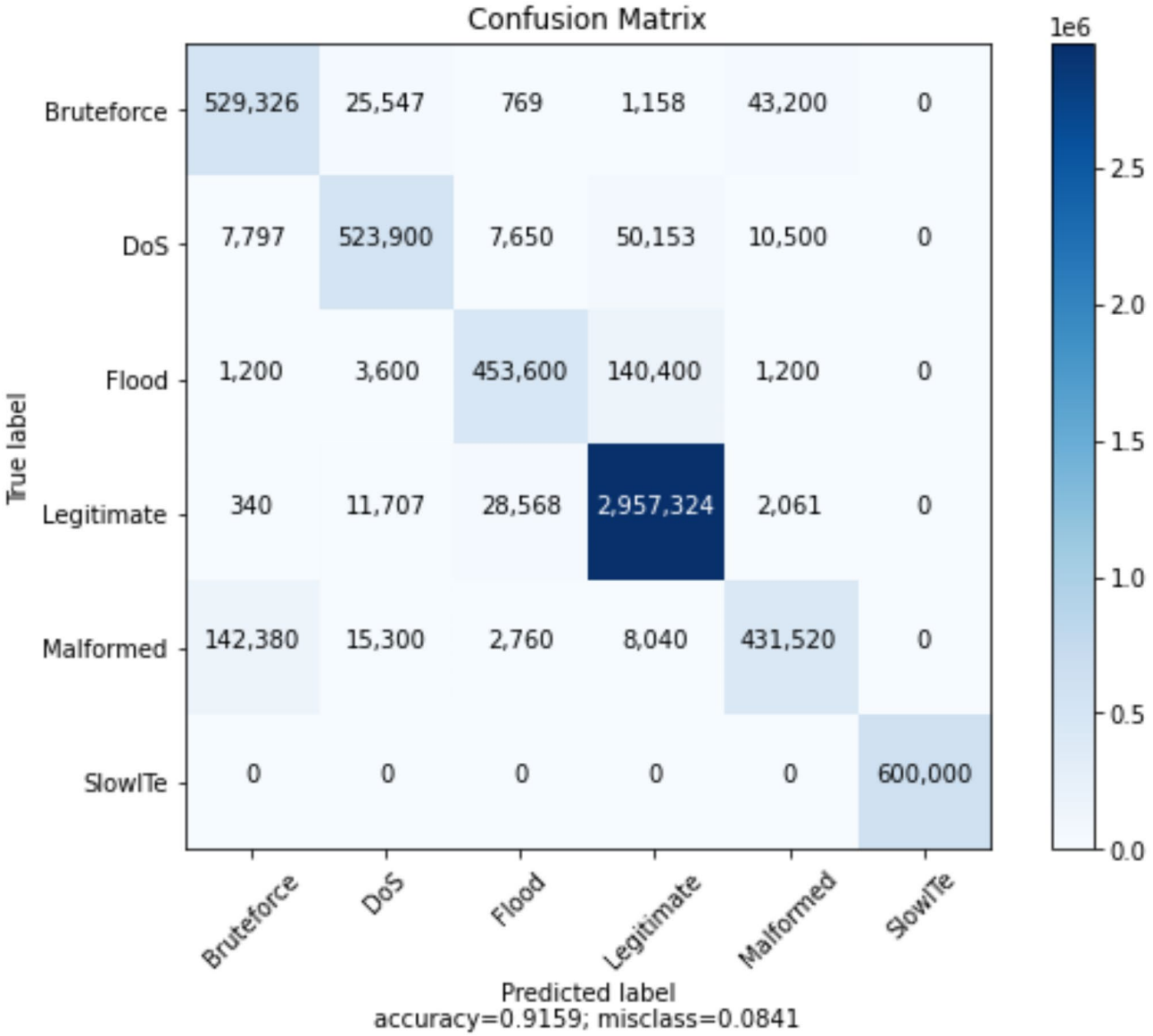


Fig. 9. Decision tree model confusion matrix.

Class	Recall	Precision	F1-Score	Accuracy
Brute Force	0.8822	0.7772	0.8264	0.9159
Dos	0.8732	0.9032	0.8879	
Flood	0.756	0.9194	0.8297	
Legitimate	0.9858	0.9367	0.9606	
Malformed	0.7192	0.8834	0.7929	
SlowITe	1	1	1	

Table 8. Decision tree model performance metrics for each class.

than naïve bayes model. Similarly, the proposed model average recall value is 0.933 which is 6.3% better than random forest and decision tree models, approximately 8.1% better than Neural network and MLP models. The proposed model attained recall is approximately 10.6% better than gradient boost and 27% better than naïve bayes model.

Figure 12 depicts the comparative analysis of F1-Score and accuracy metrics for all the models and it can be observed that the performance of proposed FL-CNN is better than the other algorithms. The proposed model average F1-score is 0.9404 which is 5.4% better than random forest and decision tree models, approximately

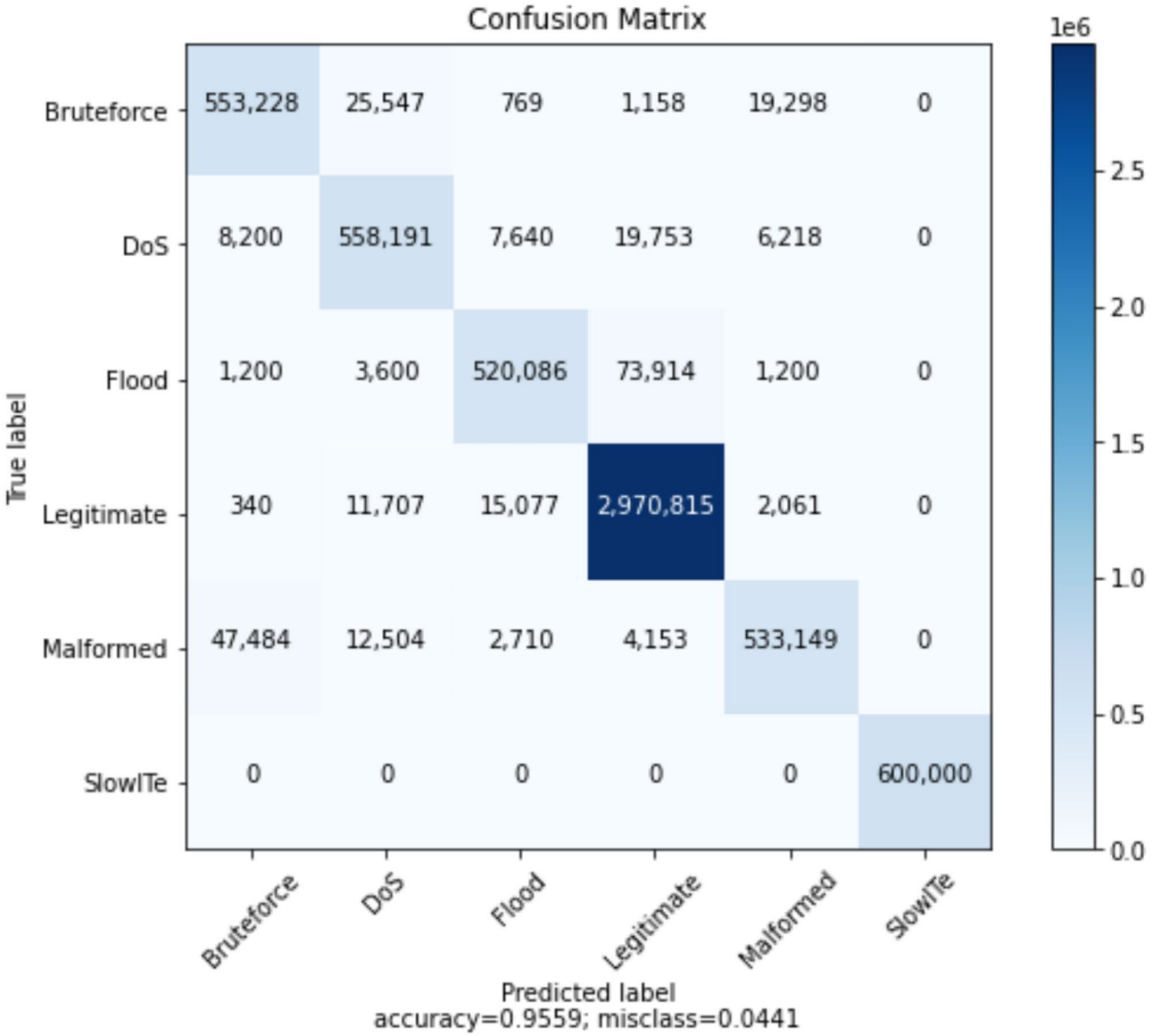


Fig. 10. Proposed FL-CNN model confusion matrix.

Class	Recall	Precision	F1-Score	Accuracy
Brute Force	0.922	0.9063	0.9141	0.9559
Dos	0.9303	0.9127	0.9214	
Flood	0.8668	0.952	0.9074	
Legitimate	0.9903	0.9678	0.9789	
Malformed	0.8886	0.9488	0.9177	
SlowITe	1	1	1	

Table 9. Proposed FL-CNN model performance metrics for each class.

7.4% better than Neural network and 7.75% better than MLP model. The proposed model attained F1-score is approximately 9.8% better than gradient boost and 20% better than naïve bayes model. Similarly, the proposed model accuracy value is 0.9559 which is 4% better than random forest and decision tree models, approximately 5.14% better than Neural network and 5.21% better than MLP model. The proposed model accuracy is approximately 7% better than gradient boost and 31% better than naïve bayes model. Due to poor feature selection and classification performance the detection performance of naïve bayes is greatly reduced compared to all other algorithms.

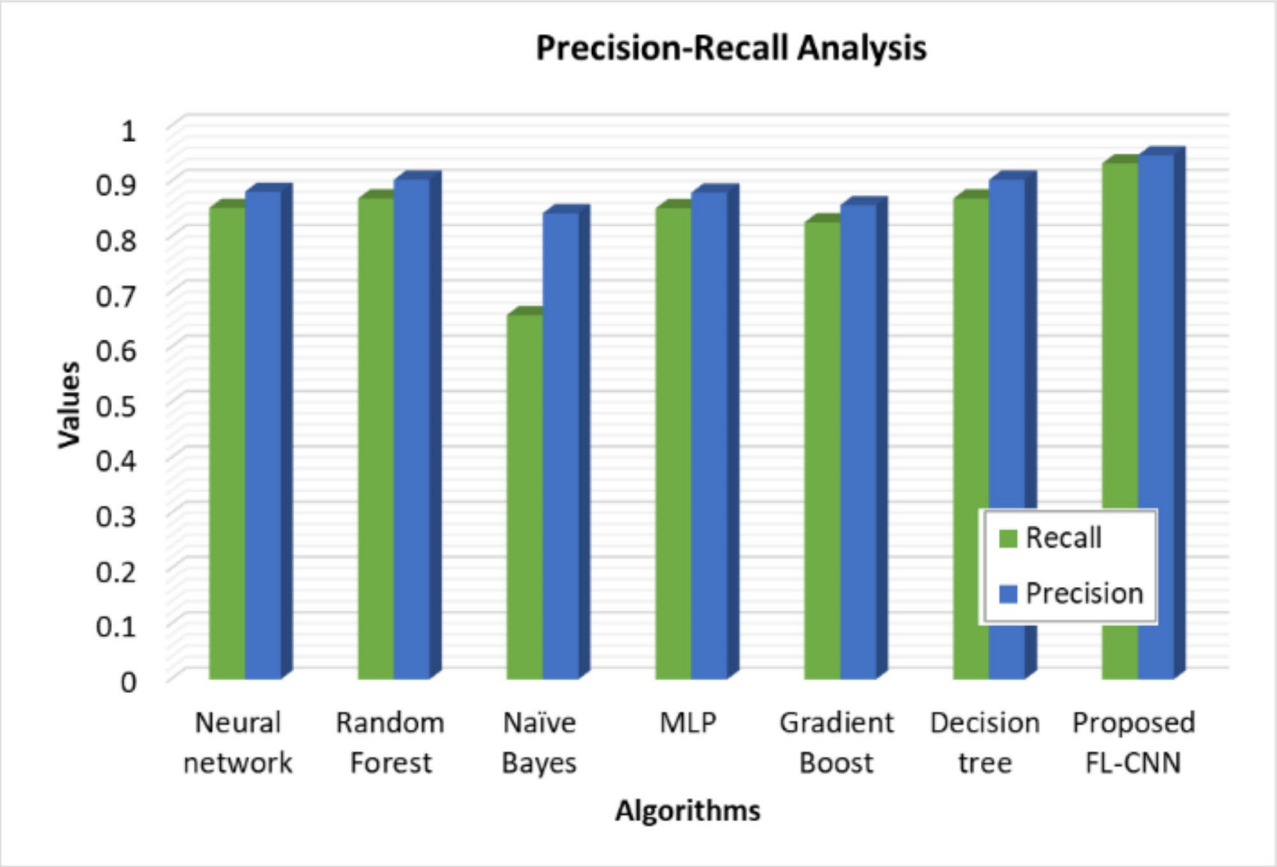


Fig. 11. Comparative analysis of precision and recall metrics.

Table 10 provides the overall performance metrics values of all the algorithms used in the experimentation. The proposed FL-CNN model has attained the maximum values for all the metrics, which indicates its better detection performance. Due to the federated learning procedure, and global model-based parameter update the detection performance is greatly improved in the IoT instances which improves the overall security of the network.

Conclusion

In this research work, an optimal federated learning-based IDS for IoT environment is presented. The concept of sharing the network parameters and developing a global model to ensure maximum accuracy in intrusion detection is incurred through the federated learning procedure. The presented IDS has optimal feature selection module which selects the optimal features from the network data to detect the intrusions. The deep learning classifier detects different types of intrusions based on the learning parameters provided by the FL global model. The detection performance is improved using the federated learning procedure compared to traditional machine learning based intrusion detection models. Experimental analysis which utilizes MQTT dataset validates the performances of proposed FL-CNN and conventional machine learning models. The maximum detection accuracy was attained by the proposed FL-CNN in detecting different types of attacks. Though the proposed model better detection performance it has few limitations. If the local model gets affected and sends wrong parameters for model aggregation due to malicious data in the local training, then it will affect the global model accuracy. Also, the optimization algorithm slightly increases the computational complexity in IoT systems. To overcome this in future, computationally efficient federated learning techniques can be developed which suitable resource-constrained IoT devices such as low-power edge devices or sensors.

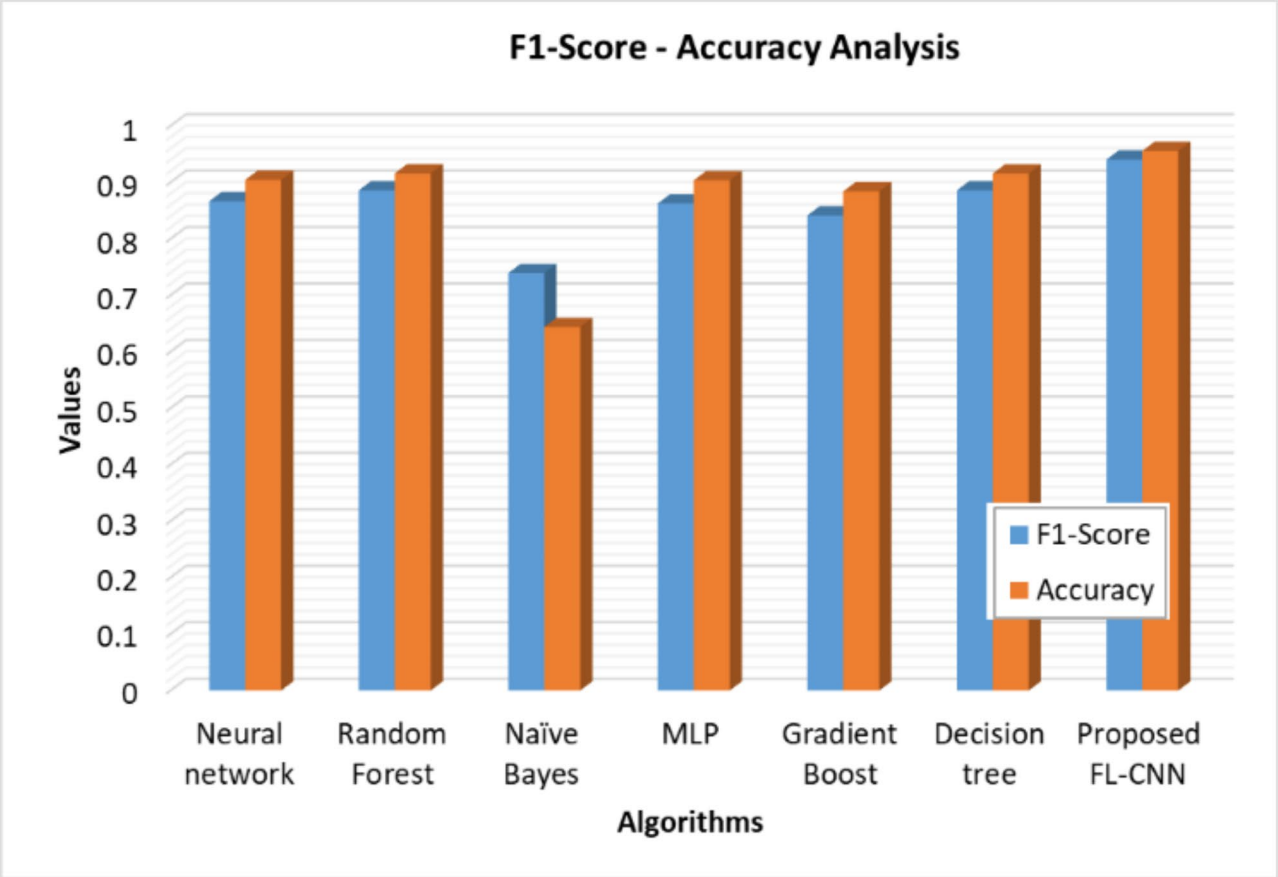


Fig. 12. Comparative analysis of F1-Score and Accuracy metrics.

S.No	Methods	Recall	Precision	F1-Score	Accuracy
1	Neural network[29]	0.8523	0.8818	0.8668	0.9045
2	Random Forest[29]	0.8694	0.9034	0.8861	0.9160
3	Naïve Bayes[29]	0.6588	0.8427	0.7395	0.6439
4	MLP[29]	0.8519	0.8803	0.8629	0.9038
5	Gradient Boost[29]	0.8264	0.8575	0.8416	0.8842
6	Decision tree[29]	0.8694	0.9033	0.8860	0.9159
7	**Proposed FL-CNN**	**0.9330**	**0.9479**	**0.9404**	**0.9559**

Table 10. Overall performance metrics for each model.

Data availability

Data is provided within the manuscript.

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References

1. Bimal, G., Danda, B. & Rawat Recent advances on federated learning for cybersecurity and cybersecurity for federated learning for internet of things. *IEEE Internet Things J.* **9** (11), 8229–8249 (2022).
2. Enrique Mármol Campos, J. L., Hernández-Ramos, J. B., Bernabé, G., Baldini, A. & Skarmeta Evaluating Federated Learning for intrusion detection in Internet of Things: Review and challenges, Computer Networks, vol. 203, pp. 1–16, (2022).
3. Hamdi, N. Federated learning-based intrusion detection system for internet of things. *Int. J. Inf. Secur.* **22**, 1937–1948 (2023).
4. Sawsan Abdul Rahman. Hanine Tout; Chamseddine Talhi; Azzam Mourad, internet of things intrusion detection: centralized, On-Device, or federated learning. *IEEE Netw.* **34** (6), 310–317 (2020).
5. Rekha, V., Manoharan, S., Hemalatha, J., Saravanan, D. & R. & Deep learning models for multiple face mask detection under a complex big data environment. *Procedia Comput. Sci.* **215**, 706–712 (2022).

6. Manoharan, J. S. Double attribute-based node deployment in wireless sensor networks using novel weight-based clustering approach. *Sadhana* **47** (3), 1–11 (2022).
7. Dapeng Man, F. et al. Intelligent Intrusion Detection Based on Federated Learning for Edge-Assisted Internet of Things, Security and Communication Networks, vol. 2022, pp.1–11. (2021).
8. Muhammad Shafiq, Z., Tian, A. K., Bashir, A., Jolfaei, X. & Yu Data mining and machine learning methods for sustainable smart cities traffic classification: A survey. *Sustainable Cities Soc.* **60**, 102177 (2020).
9. Muhammad Shafiq, Z., Tian, Y., Sun, X., Du, M. & Guizani Selection of effective machine learning algorithm and Bot-IoT attacks traffic identification for internet of things in smart City. *Future Generation Comput. Syst.* **107**, 433–442 (2020).
10. Rashid, M. M. et al. A federated Learning-Based approach for improving intrusion detection in industrial internet of things networks. *Network* **3** (1), 158–179 (2023).
11. Ruijie, Z. & Gui, G. Yijun Wang; Zhi Xue; Tomoaki Ohtsuki; Bamidele Adebisi; Semisupervised Federated-Learning-Based Intrusion Detection Method for Internet of Things, IEEE Internet of Things Journal, vol. 10, no. 10, pp. 8645–8657, (2023).
12. Xabier Sáez-de-Cámara, Flores, J. L., Arellano, C., Urbieta, A. & Zurutuza, U. Clustered federated learning architecture for network anomaly detection in large scale heterogeneous IoT networks. *Computers Secur.* **131**, 1–20 (2023).
13. Francisco Lopes de Caldas Filho, Soares, S. C. M. & Oroski, E. Robson de Oliveira Albuquerque, Rafael zerbini Alves Da Mata, Fábio Lúcio lopes de mendonça, Rafael Timóteo de Sousa júnior, botnet detection and mitigation model for IoT networks using federated learning. *Sensors* **23** (14), 1–35 (2023).
14. Pedro Ruzafa-Alcázar; Pablo Fernández-Saura; Enrique Mármol-Campos; Aurora González-Vidal; José L. Hernández-Ramos; Jorge Bernal-Bernabe, Antonio, F. & Skarmeta Intrusion Detection Based on Privacy-Preserving Federated Learning for the Industrial IoT, IEEE Transactions on Industrial Informatics, vol. 19, no. 2, pp. 1145–1154, (2023).
15. Vasiliki Kelli, V. et al. IDS for industrial applications: A federated learning approach with active personalization. *Sensors* **21** (20), 1–17 (2021).
16. Zhang, J., Luo, C., Marcus, C. & Min, G. Federated learning for distributed IIoT intrusion detection using transfer approaches. *IEEE Trans. Industr. Inf.* **19** (7), 8159–8169 (2023).
17. Idrissi, M. J. et al. Fed-ANIDS: federated learning for anomaly-based network intrusion detection systems. *Expert Syst. Appl.* **234**, 1–11 (2023).
18. Hajj, S. et al. Cross-Layer federated learning for lightweight IoT intrusion detection systems. *Sensors* **23** (16), 1–26 (2023).
19. Jonathas, A. et al. F-NIDS — A network intrusion detection system based on federated learning. *Comput. Netw.* **236**, 1–14 (2023).
20. Othmane Friha, M. A. et al. FELIDS: Federated learning-based intrusion detection system for agricultural Internet of Things, Journal of Parallel and Distributed Computing, vol.165, pp. 17–31, (2022).
21. Kavitha, M. et al. Smart Environment and Plant Disease Monitoring with Analysis System in Agricultural Field, 2023 IEEE International Conference on ICT in Business Industry & Government (ICTBIG), Indore, India, 08–09 December 2023.
22. Viraaji Mothukuri; Prachi Khare, Reza, M. & Parizi Seyedamin Pouriyeh; Ali Dehghantanha; Gautam Srivastava, Federated-Learning-Based anomaly detection for IoT security attacks. *IEEE Internet Things J.* **9** (4), 2545–2554 (2022).
23. Pan, Z. C. Intrusion Detection for Wireless Edge Networks Based on Federated Learning, IEEE Access, vol. 8, pp. 217463–217472, (2020).
24. Asuvaran, A. & Senthilkumar, S. Low delay error correction codes to correct stuck-at defects and soft errors, 2014 International Conference on Advances in Engineering and Technology (ICAET), 02–03 May (2014).
25. Muhammad Shafiq; Zhihong Tian; Ali Kashif Bashir. Xiaojiang Du; Mohsen Guizani, CorrAUC: A malicious Bot-IoT traffic detection method in IoT network using Machine-Learning techniques. IEEE Internet Things J. **8**, 5, pp. 3242–3254, (2021).
26. Muhammad Shafiq, Z., Tian, A. K., Bashir, X., Du, M. & Guizani IoT malicious traffic identification using wrapper-based feature selection mechanisms. *Computers Secur.* **94**, 101863, (2020).
27. Zhao, R., Yin, Y., Shi, Y. & Xue, Z. Intelligent intrusion detection based on federated learning aided long short-term memory. *Phys. Commun.* **42**, 1–12 (2020).
28. Shafiq, M., Tian, Z., Bashir, A. K., Du, X. & Guizani, M. CorrAUC: A malicious Bot-IoT traffic detection method in IoT network using Machine-Learning techniques. *IEEE Internet Things J.* **8** (5), 3242–3254. https://doi.org/10.1109/JIOT.2020.3002255 (2021).
29. Shafiq, M., Tian, Z., Bashir, A. K., Du, X. & Guizani, M. IoT malicious traffic identification using wrapper-based feature selection mechanisms. *Computers Secur.* **94**. https://doi.org/10.1016/j.cose.2020.101863
30. https://www.kaggle.com/datasets/cnrieiit/mqttset.
31. Ivan Vaccari, G., Chiola, M., Aiello, M., Mongelli, E. & Cambiaso MQTTset, a New Dataset for Machine Learning Techniques on MQTT, Sensors, vol. pp.1–17, 2020. (2020).

Author contributions
All the authors are contributed this research work draft this manuscript.

Declarations

Competing interests
The authors declare no competing interests.

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